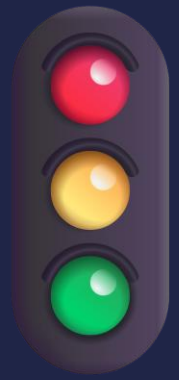
Master 1 RSD · Modeling & Simulation



Network Queueing

Why does everything have to wait?

Modeling & Simulation of Networks · Master 1 RSD
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Queues are everywhere



IT: the router

Packets arrive faster than the link can send, so they wait in the buffer.



Healthcare: the ER

Patients arrive at random. Doctors serve them one by one.



Factory: the line

Parts pile up in front of the slowest machine.

Same story: arrive → wait → get served → leave.

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Reading the code: A / B / c

A

Arrivals

M = random arrivals (Poisson), λ per second.

B

Service time

M = random duration (exponential), average $1/\mu$.

c

Servers

How many cashiers, doctors or CPUs work in parallel.

M/M/1 = one machine · M/M/2 = two doctors · M/M/c = c servers, one shared line

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One server: M/M/1

A router gets $\lambda = 40$ packets/s and sends $\mu = 50$ packets/s.

Busy ratio

$$\rho = \lambda/\mu = 0.8$$

Must stay below 1

Time in system

$$W = 1/(\mu-\lambda) = 0.1 \text{ s}$$

Wait + service

Little's law

$$L = \lambda W = 4$$

Packets inside, on average

🚗 🚗 🚗 **When ρ gets close to 1, waiting explodes, like rush-hour traffic.**

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Your queue toolkit

Same router as before: $\lambda = 40/s$, $\mu = 50/s$, so $\rho = 0.8$.

Little's law

$$L = \lambda \cdot W$$

Works for any queue. 40 × 0.1 s = 4 packets inside.

Wait + service

$$W = W_q + 1/\mu$$

0.1 s = 0.08 s waiting + 0.02 s being served.

Waiting only

$$W_q = \rho/(\mu-\lambda)$$

= 0.08 s. Packets in the line: $L_q = \lambda \cdot W_q = 3.2$.

Chance of n inside

$$P(n) = (1-\rho)\cdot\rho^n$$

$P(0) = 0.2$, so the server is idle 20% of the time.

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Exercise 1: the clinic 🏥

One doctor. Patients arrive at random, **6 per hour**. A consultation lasts **8 minutes** on average.

1. What is μ per hour, and what is ρ?
2. Average time a patient spends in the clinic?
3. Average number of patients inside?
4. A nurse cuts the visit to 5 min. What changes?

⏱ **work in pairs**

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When the buffer is full

M/M/1/K: room for only K jobs in total. Extra arrivals are dropped. Router with $\rho = 0.8$:

Buffer K = 5

≈ 8.9% lost

Almost 1 packet in 11 is dropped.

Buffer K = 10

≈ 2.3% lost

Doubling the buffer cuts losses by about 4x.

Loss chance: $P_K = (1-\rho)\cdot\rho^K / (1-\rho^{(K+1)})$

🏭 Full buffer: the machine upstream stops. 🏥 No free bed: the ambulance goes elsewhere.

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Share the line, win big



3 × M/M/1

Each doctor has their own line. One sits idle while another line overflows.



1 × M/M/3

One shared line. Next free doctor takes the next patient. Nobody idles.

Datacenter version: one load balancer in front of 3 servers beats 3 separate queues.

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How many servers do we buy?

λ = 150 req/s · μ = 50 req/s per server · goal: wait Wq < 20 ms

💥

3 servers

ρ = 1.00

Wq = ∞

Collapse

✅

4 servers

ρ = 0.75

Wq ≈ 10 ms

Sweet spot

💸

5 servers

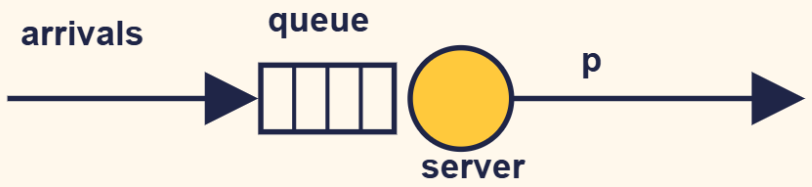
ρ = 0.60

Wq ≈ 2.4 ms

Overkill

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Queueing networks: draw them!



box = waiting line, circle = server, arrow = job moves with pr

Open network

🏥 Hospital

Patients come from outside and leave. Arrival rate γ is fixed.

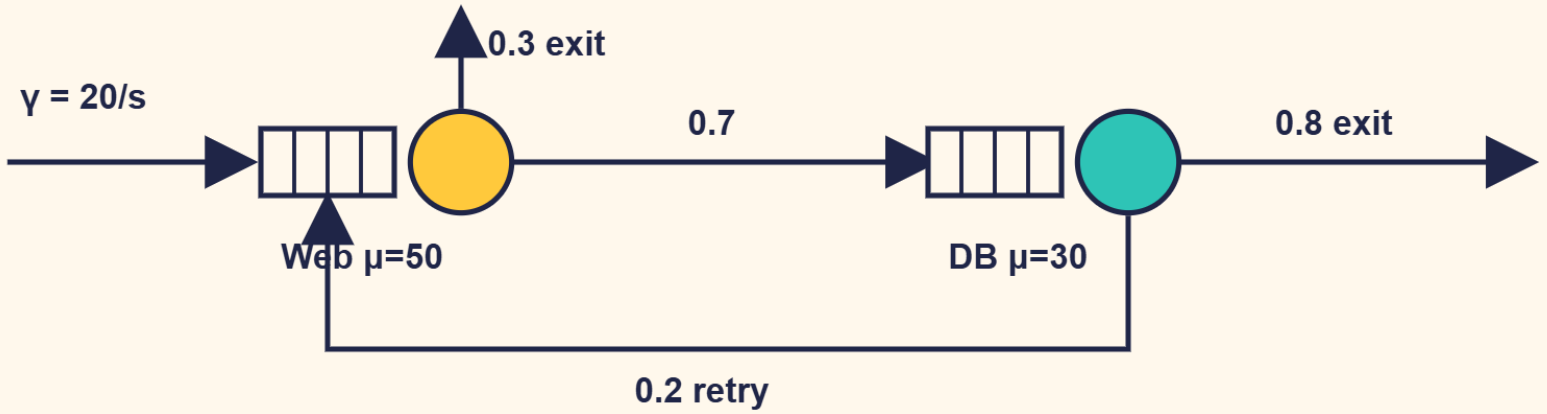
Closed network

🏭 Pallets on a loop

A fixed number N of jobs circulate. Nobody enters or leaves.

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Open network example: web shop 🛒



Traffic equations: **$\lambda 1 = 20 + 0.2 \cdot \lambda 2$** and **$\lambda 2 = 0.7 \cdot \lambda 1$**
So $\lambda 1 \approx 23.3/s$ and $\lambda 2 \approx 16.3/s$

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Jackson's trick

In an open network, each node behaves like its **own M/M/1**. Just use its own λ.

Web server

$\rho \approx 0.47$

About 0.87 requests inside.

Database

$\rho \approx 0.54$

About 1.19 requests inside.

Whole shop

$W \approx 103$ ms

Total inside ≈ 2.06, so $W = L/\gamma$.

Total time = sum over nodes of (L_i) divided by outside arrivals γ.

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Exercise 2: the factory 🏭

Parts arrive at **10/min**, go to **Cutting (μ=20)**, then **Painting (μ=15)**. After painting, **20%** go back to cutting for rework. The rest ship.

1. Draw the network and write the traffic equations.
2. Find ρ of each machine. Which is the bottleneck?
3. Average parts in the factory? Average time?
4. Rework rises to 40%. What happens?

⏱ **draw first!**

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From theory to your TP

1

Random numbers

Your LCG gives U in [0,1[. A bad LCG gives bad traffic.

2

Make arrivals

Gap between packets: T = -ln(1-U)/λ

3

Compare

Simulate the queue, then check your average W against 1/(μ-λ).

Same idea in the TP: LCG → histogram → does it look uniform?

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